George Raafat

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Software Engineer

Backend | DevOps | ERPNext & Frappe

Summary

Software Engineer with experience developing production ERP systems and scalable backend applications. Skilled in REST APIs, database design, Docker, and collaborative development.

Education

Cairo University — **Faculty of Computers & Artificial Intelligence**
B.Sc. in Computer Science - GPA: 3.81/4.0 — *10/2022 – 06/2026*

Experience

Appy Innovate — **Remote**
Software Engineer (Backend & ERPNext) — *04/2026 – Present*

- Built a catch-weight module integrating hardware scales and raw ZPL barcode printing, automating POS scans and reducing manual operation time by ∼80%.
- Implemented Phase 2 ZATCA e-invoicing to automatically submit ∼75 daily production invoices and generate comprehensive VAT reports.
- Customized POS logic to enforce conditional pricing and scanning rules, and created multi-user acknowledgment workflows for secure stock transfers.
- Discovered and patched a core ERPNext bug, resolving inaccurate valuation and COGS calculations.
- Set up deployment infrastructure using Proxmox VMs and Docker, automating wildcard SSL generation via Nginx Proxy Manager and Cloudflare DNS.
- Deployed and maintained self-hosted enterprise ecosystems (Keycloak with OAuth2-proxy for SSO, Huly, Moodle) and integrated local LLMs (Qwen3 via Ollama) to power AI-assisted developer workflows.

Aehter — **Remote**
Freelance Software Engineer (ERPNext & DevOps) — *07/2026 – 09/2026*

- Consulted with stakeholders to align client requirements with development, delivering a fully white-labeled Frappe HR ERP system.
- Engineered a custom Disciplinary Action module (reports, investigations, appeals) alongside dynamic analytics reports to track HR metrics and workflows.
- Localized employee data models for Egyptian compliance.
- Managed end-to-end Docker deployment on a Hostinger VPS using Cloudflare and NPM.

NCI (via ALX Ventures) — **Remote**
Backend Developer — *09/2024 – 11/2024*

- Built and deployed a scalable API for an inventory management SaaS (Nest.js, MongoDB).
- Implemented multi-tenant database architecture to streamline client data management.

Training & Certifications

Banque Misr — **Hybrid**
Backend Development Intern — *07/2025 – 08/2025*

- Developed secure backend applications using Spring Boot, JWT, MySQL, and MongoDB.
- Implemented automated testing and containerized applications with Docker.

ALX Africa — **Remote**
Software Engineering Intern — *02/2023 – 04/2024*

- Completed a 12-month software engineering program specializing in backend development.
- Developed full-stack web applications while designing relational and NoSQL databases using modern backend frameworks.
- Applied Linux, Git, agile workflows, and test-driven development in collaborative software projects.

Projects

SmartGallery (Graduation Project) *10/2025 – 06/2026*

- Led a five-member team developing an AI-powered Android gallery application.
- Designed the MVVM architecture and implemented semantic search and image indexing.
- Converted MobileCLIP2-S0 to ONNX and reduced the text encoder from 224 MB to 61 MB.
- Enabled fully offline semantic, face, and OCR search, indexing 1,000 images in ∼15 minutes.

Learning Management System *12/2024 – 12/2024*

- Built a course management platform (Spring Boot, PostgreSQL) featuring normalized schemas and Cloudinary media integration.

Doodles *03/2024 – 04/2024*

- Developed a drawing-sharing web app (Next.js, React, Prisma, MongoDB, NextAuth).
- Implemented authentication, optimized the database, and set up image storage with Cloudinary.

Skills

Programming Languages: Python, JavaScript/TypeScript, Java, Kotlin, C, C++

Frameworks: Frappe, Flask, Django, Nest.js, Next.js, Spring Boot, Jetpack Compose

Databases: MySQL, PostgreSQL, SQLite, SQL Server, MongoDB

ORMs: Room, Prisma, JPA

DevOps & Tooling: Linux, Git, Docker, Proxmox, Nginx, Cloudflare, OpenSSL, ONNX Runtime

Web Technologies: HTML, CSS, Tailwind CSS